3. AMD Workbench

AMD AI Workbench is the end-user interface for deploying and interacting with AI models. It is accessed separately from the Resource Manager — navigate to the url exposed after installation:

For a .nip.io domain (default if installation via Digital Ocean): https://kc..nip.io For a registered domain: https://airmui.

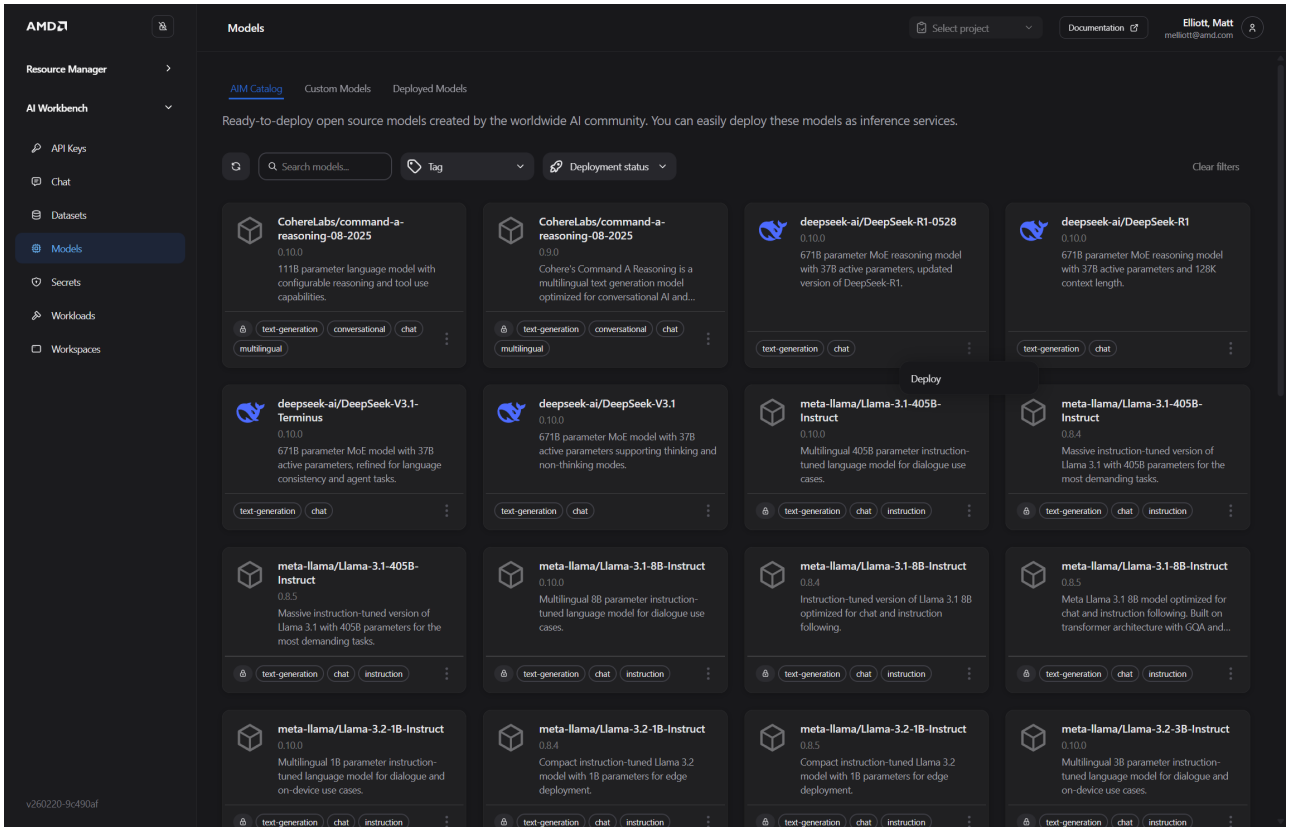
Log in with the same credentials used for the Resource Manager. Ensure you are working within the project you created in the previous section.

Finetuning

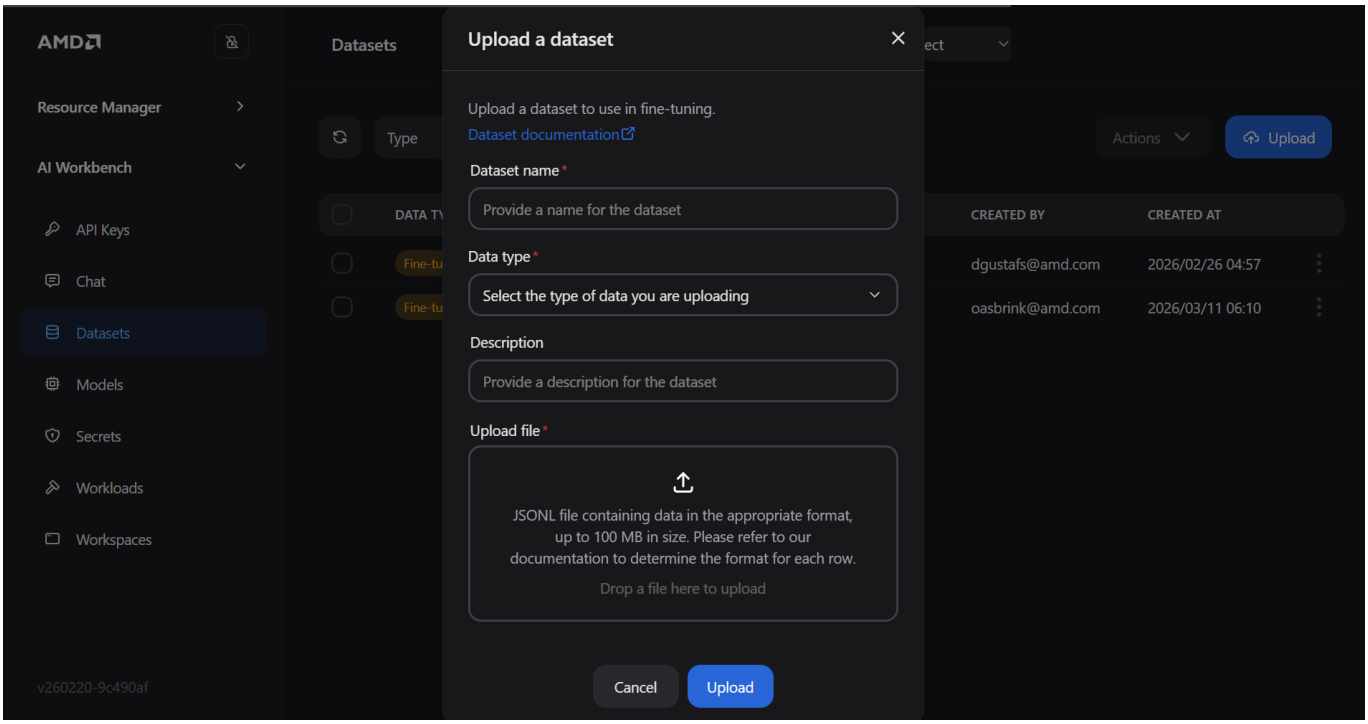
Finetuning allows you to adapt a base model to domain-specific data.

Typical Workflow

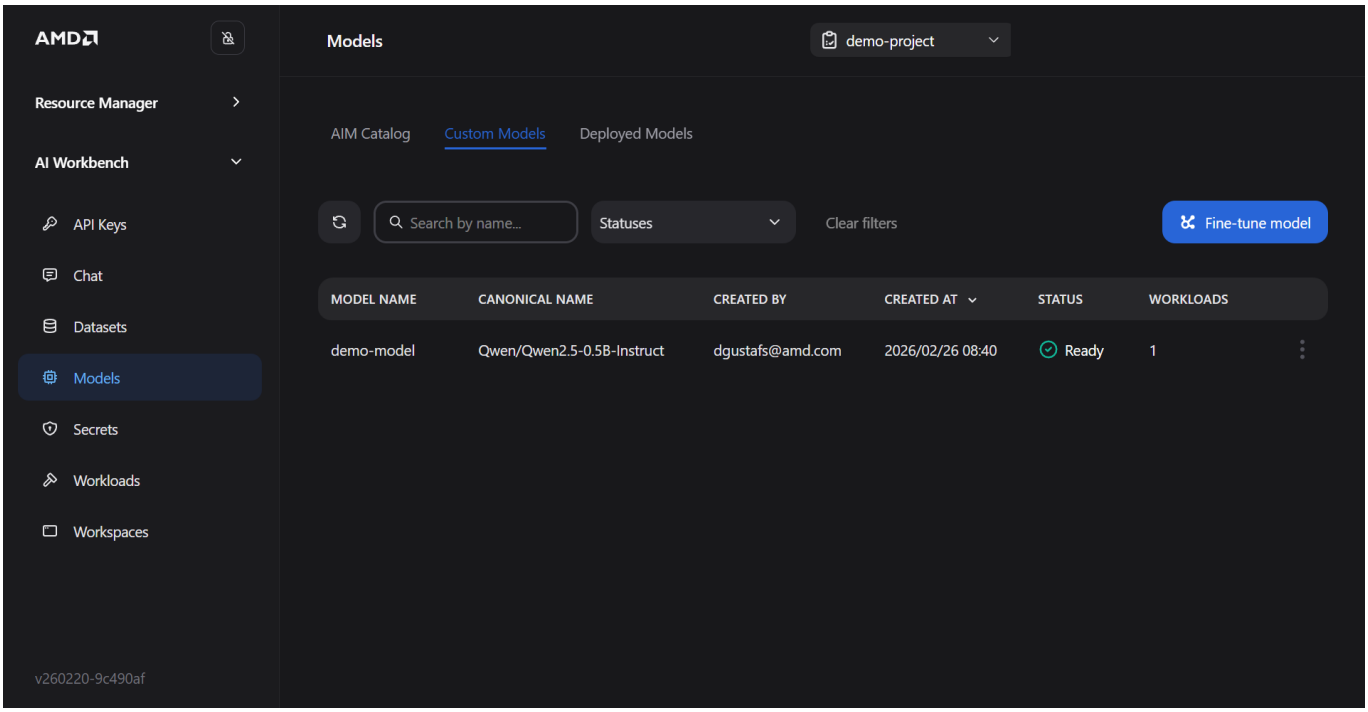
1. **Add Hugging Face token** — Required for accessing gated models and datasets



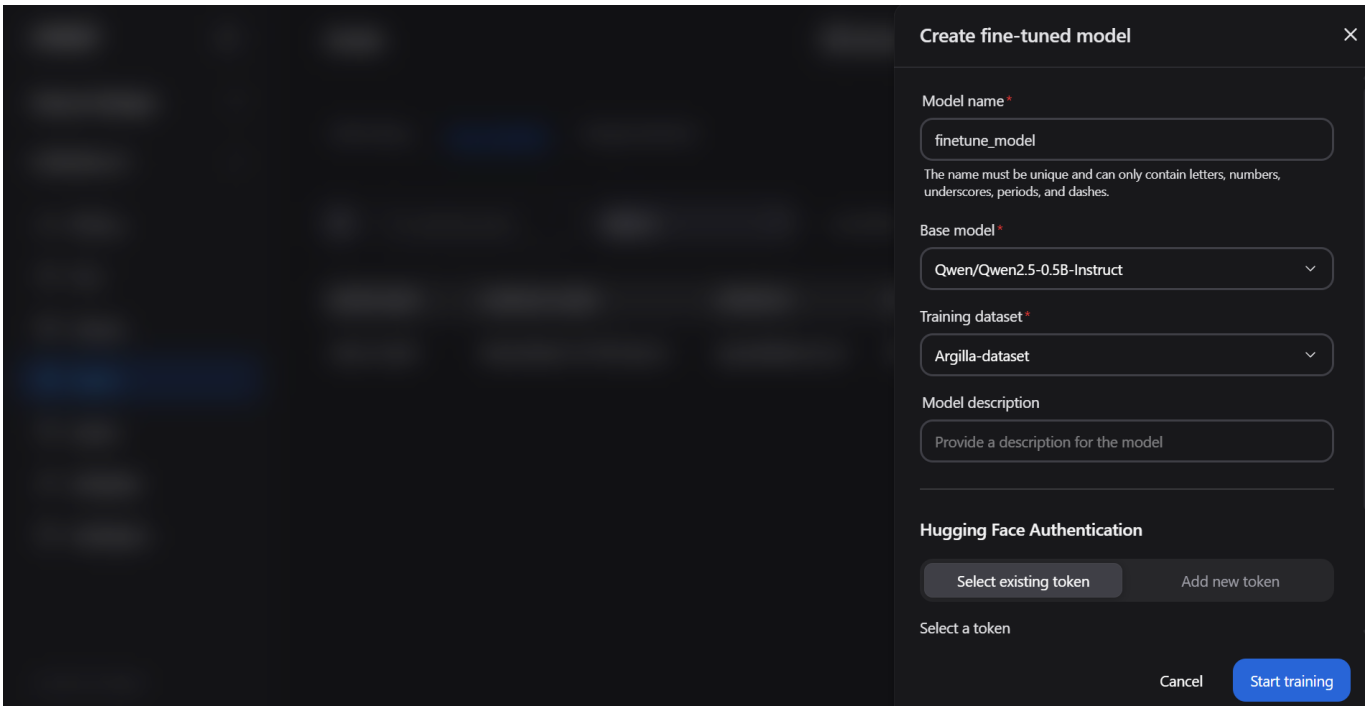
2. **Navigate to Datasets and upload training data** — In **AI Workbench**, open **Datasets** from the left navigation and click **Upload**.



3. **Create the dataset entry** — Enter a dataset name, choose the correct data type, optionally add a description, then upload your `.jsonl` file and click **Upload**.
4. **Go to Custom Models** — Open **Models** and switch to the **Custom Models** tab.

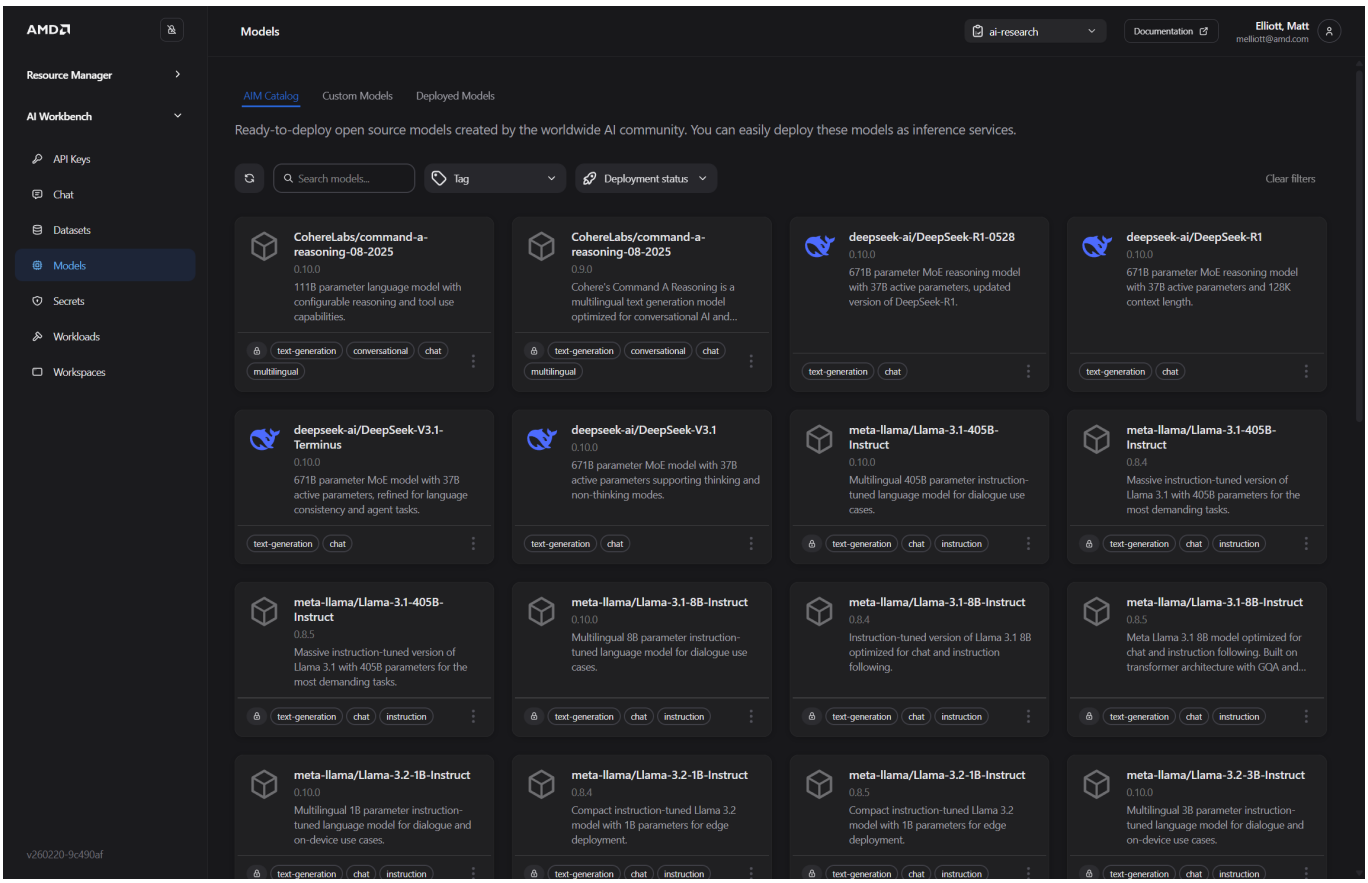


5. **Start fine-tuning** — Click **Fine-tune model**, select the base model and uploaded dataset, configure training parameters, then click **Start training**.



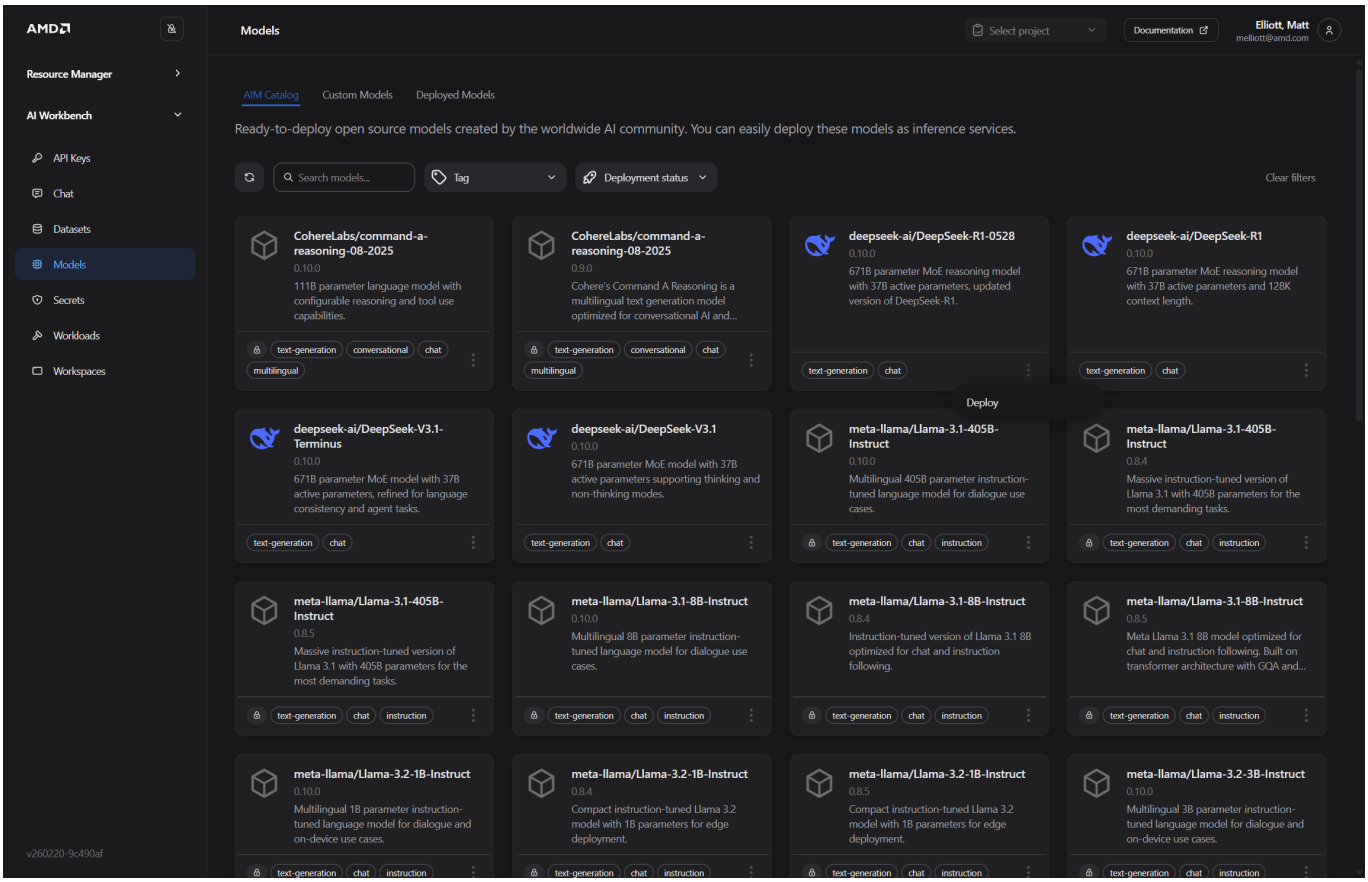
Deploy an AI Model (AIM)

> **Note:** You are in the **AMD AI Workbench** interface for this section. Ensure you have selected the correct project before proceeding.



1. Navigate to the **Models** tab to access the AIM catalog
2. Locate the model you want to deploy and click the **three-dot menu (⋮)** in the bottom-right corner of the model card

3. Select **Deploy**

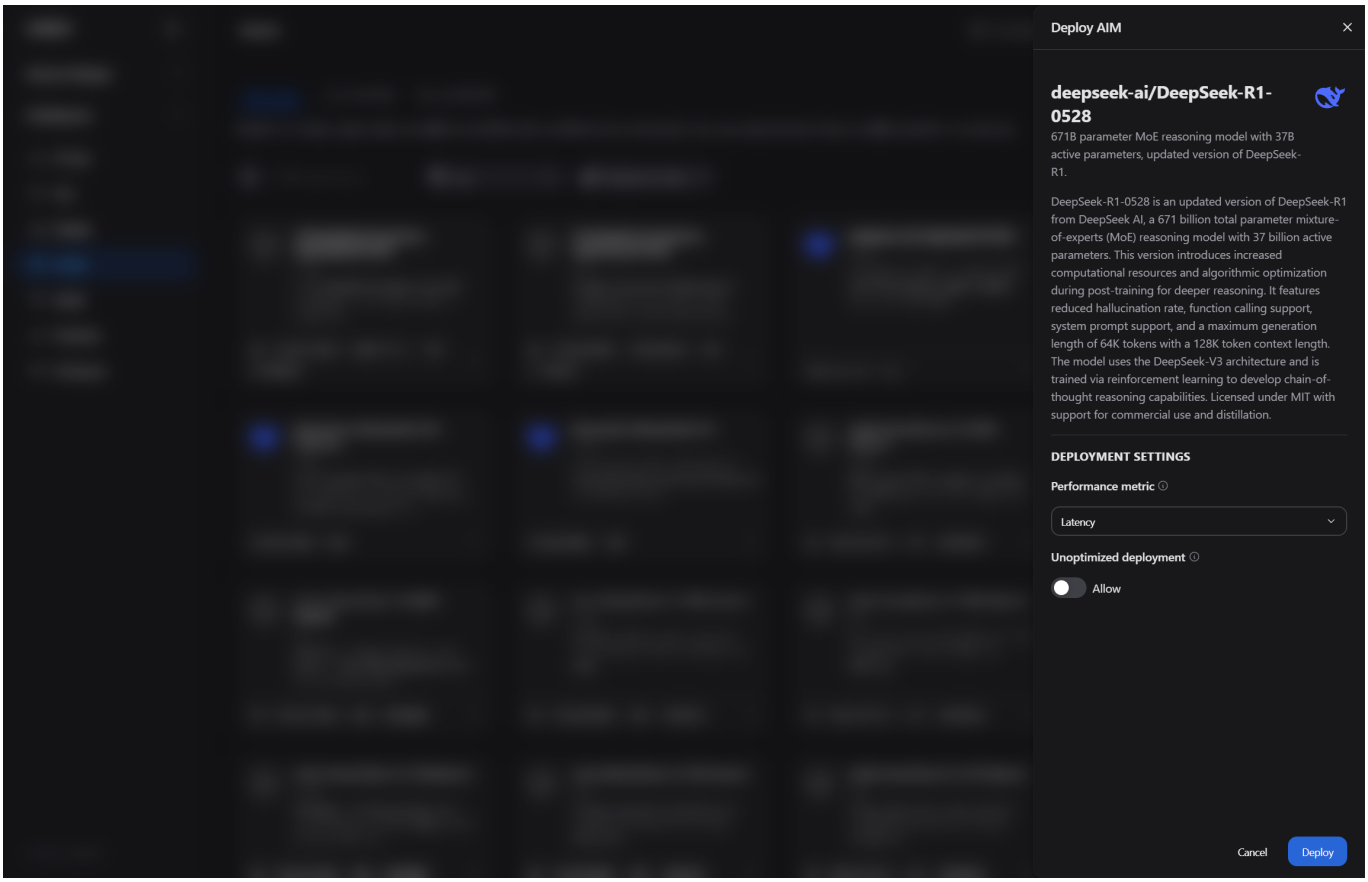


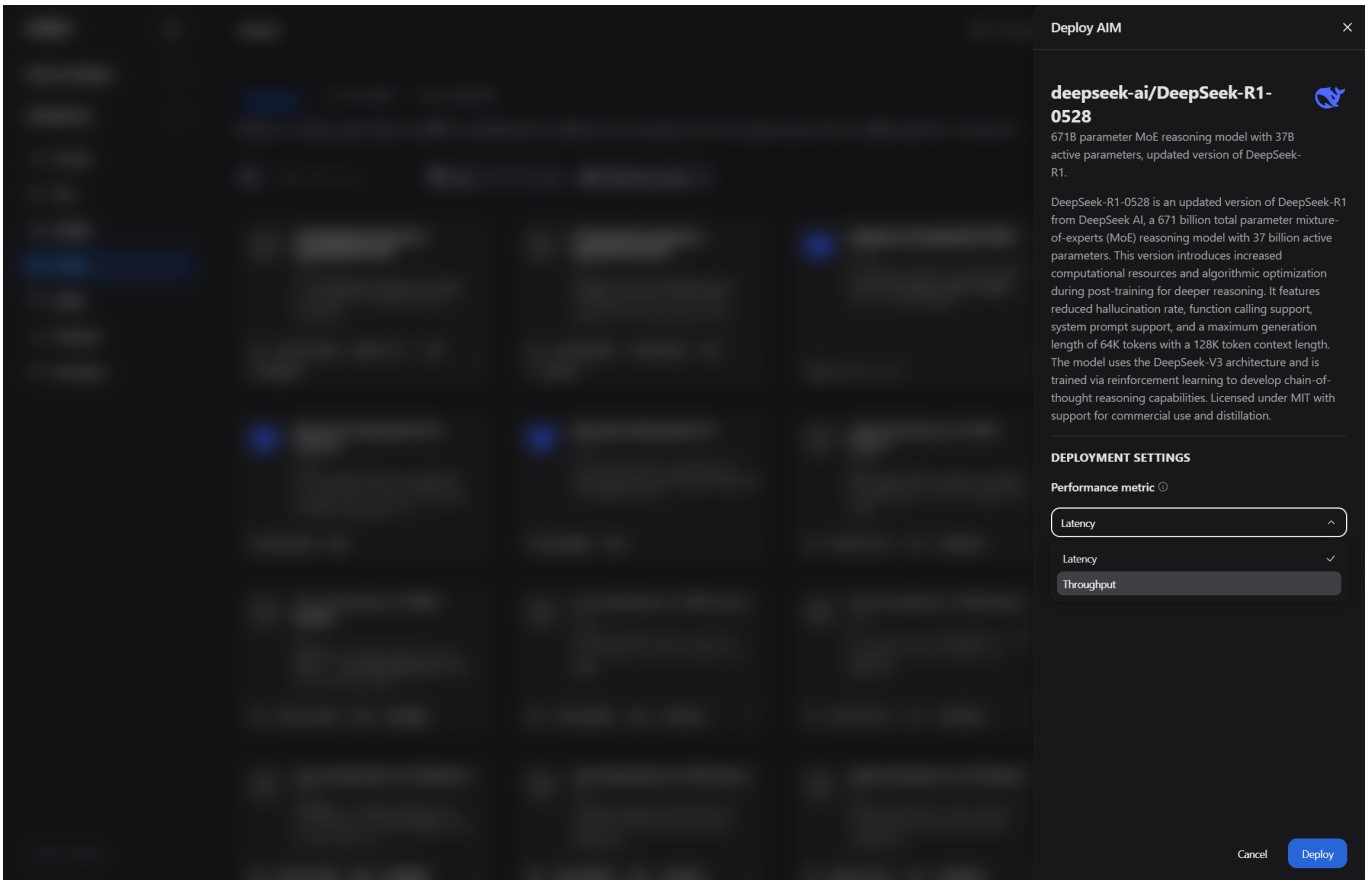
4. Configure the **Deployment Settings**:

- **Performance metric** — Select the optimization target from the dropdown:

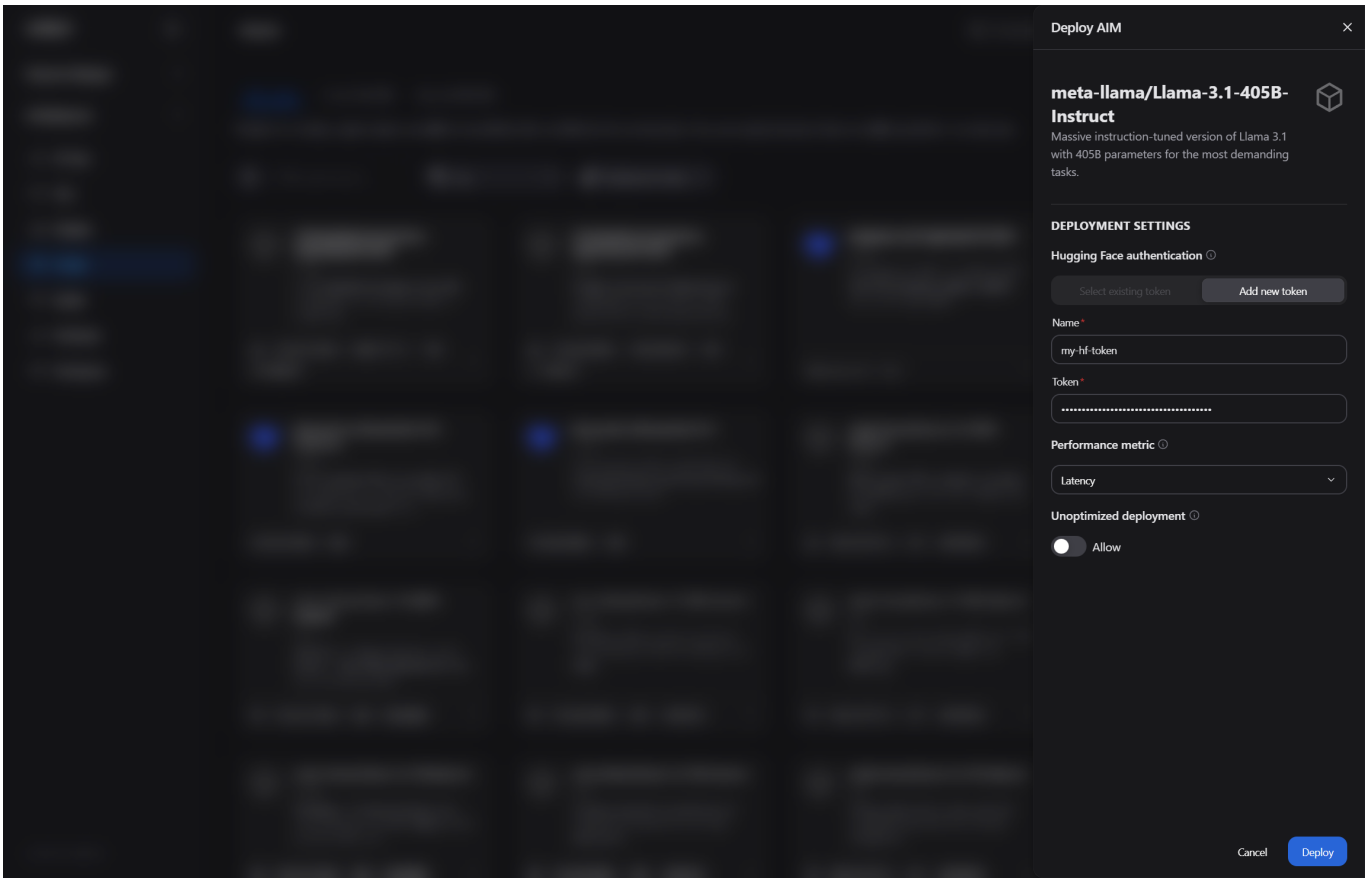
Option	When to use
Latency	When minimizing response time per request is the priority
Throughput	When maximizing sustained requests/second is the priority

- **Unoptimized deployment** — Toggle **Allow** only when deploying to hardware the AIM is not specifically optimized for. Leave this off for standard deployments.





5. If the model is gated (indicated by a lock icon — e.g., Llama family models), a **Hugging Face authentication** section appears in the panel. Either click **Select existing token** to reuse a previously stored token, or click **Add new token** and enter a name and token value directly.



6. Click **Deploy**. A confirmation message will appear indicating the workload has started.

7. **Wait for the model to become ready.** Navigate to **Workloads** to monitor deployment status. The model is ready when its status shows **Running**.

> Deployment typically takes depending on model size and cluster load.

VSCode Workspace (vLLM Benchmarking)

This section demonstrates how to use the built-in Visual Studio Code workspace to benchmark a deployed model using the `vllm bench` tool.

Launch the VSCode Workspace

1. Navigate to **Workspaces** in the left sidebar
2. Click **View and deploy** next to the Visual Studio Code workspace entry
3. Once deployed, click the **Launch** button to open VSCode in your browser

Get the Model Endpoint

Before running the benchmark, retrieve the internal endpoint of your deployed model:

1. Navigate to the **Models** tab

2. On the deployed model card, click the **three-dot menu (⋮)** and select **Connect**

3. Copy the **Internal URL** — this is the endpoint used within the cluster

> If accessing from outside the cluster (e.g., from your local machine), use the **External URL** together with an API key instead.

Run the Benchmark

Open a terminal in the VSCode workspace and run the following setup commands:

```
python --version          # Verify Python is available

python -m venv venv       # Create a virtual environment
source venv/bin/activate  # Activate it

pip install vllm          # Install the vllm benchmarking tool
```

Then configure and run the benchmark. Replace the placeholder values with those for your deployment:

```
NUM_PROMPTS=<number-of-prompts>
CONC=$((NUM_PROMPTS * 10))   # Sets concurrency to 10x the prompt count — adjust
as needed
INPUT_LEN=<input-token-length>
OUTPUT_LEN=<output-token-length>
BASE_URL="<your-internal-url>"
ENDPOINT="/v1/chat/completions"
MODEL="<your-model-name>"

vllm bench serve \
  --ignore-eos \
  --backend openai-chat \
  --base-url "${BASE_URL}" \
  --endpoint "${ENDPOINT}" \
  --model "${MODEL}" \
  --dataset-name random \
  --random-input-len ${INPUT_LEN} \
  --random-output-len ${OUTPUT_LEN} \
  --num-prompts ${NUM_PROMPTS} \
  --max-concurrency ${CONC} \
  --trust-remote-code
```

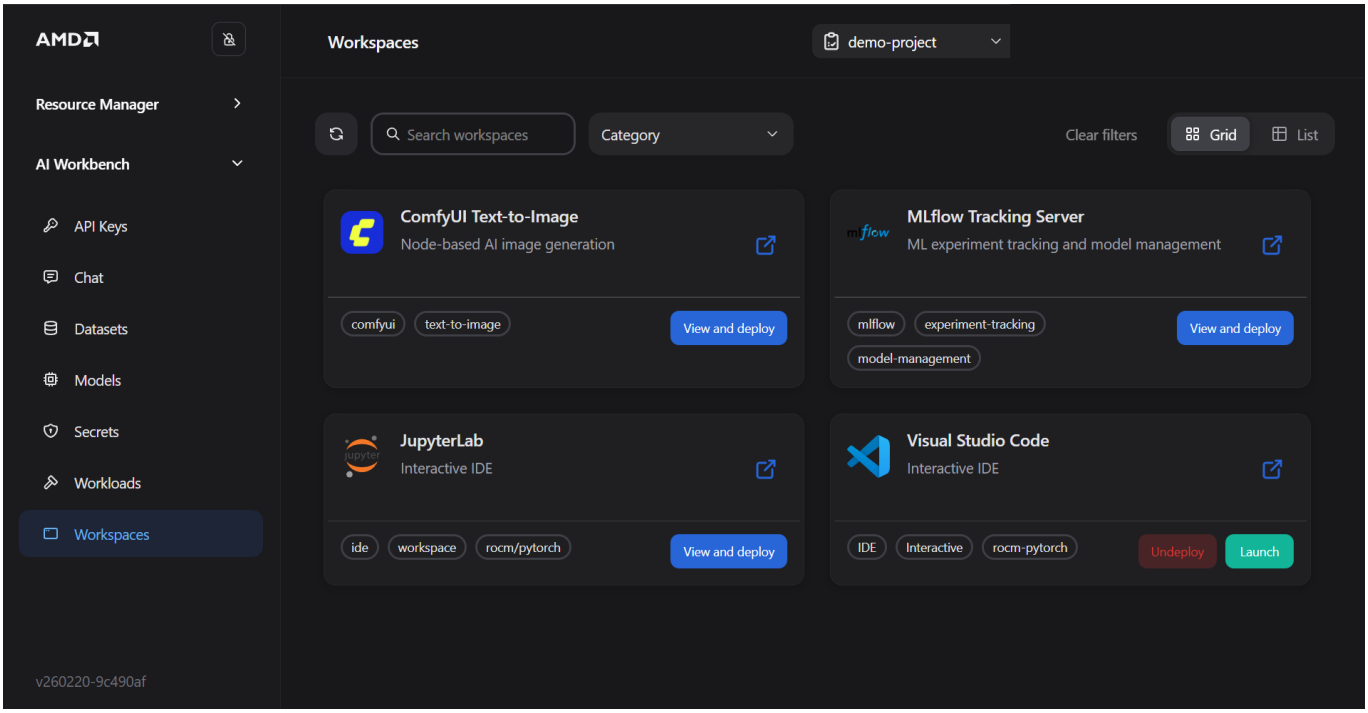
Understanding Benchmark Output

Metric	Meaning
Throughput	Total tokens processed per second across all concurrent requests
TTFT	Time to First Token — how quickly the model starts responding
Latency	End-to-end time per request
Tokens/sec	Per-request token generation rate

ComfyUI

ComfyUI provides a visual node-based interface for building and running AI pipelines, including image generation workflows.

1. Navigate to **Workspaces** in AI Workbench and locate the **ComfyUI Text-to-Image** workspace.



2. Click **View and deploy**, then allocate the appropriate number of GPUs based on workload demand.
3. After deployment is ready, click **Launch**.
4. In ComfyUI, select one of the available text-to-image templates.
5. Enter a text prompt and run the workflow to generate images.

Next: Proceed to Blueprints to deploy a solution blueprint.